

Physics Lecture 3: Thermodynamics

Introduction to the Laws of Thermodynamics

Thermodynamics is the branch of physics that deals with heat, work, and temperature, and their relation to energy, radiation, and the physical properties of matter. The behavior of these quantities is governed by the four laws of thermodynamics.

Key Concepts and Formulas

- **Zeroth Law:** If two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other. This law establishes temperature as a fundamental concept.
- **First Law:** The change in the internal energy of a system is equal to the heat added to the system minus the work done by the system on its surroundings. $\Delta U = Q - W$
 - ΔU = change in internal energy (in Joules, J)
 - Q = heat added to the system (in Joules, J)
 - W = work done by the system (in Joules, J)
- **Second Law:** The total entropy of an isolated system can never decrease over time. It can remain constant for reversible processes, but it must increase for irreversible processes. $\Delta S \geq 0$
- **Third Law:** The entropy of a system approaches a constant value as the temperature approaches absolute zero (0 K).

Mathematical Example Problem: First Law of Thermodynamics

Problem: A gas in a cylinder is heated with 400 J of energy. The gas expands, doing 150 J of work on its surroundings. What is the change in the internal energy of the gas?

Solution:

1. **Identify the given values:**
 - Heat added to the system, $Q = +400$ J (positive because heat is added).
 - Work done by the system, $W = +150$ J (positive because work is done by the system).
2. **Apply the First Law of Thermodynamics:** $\Delta U = Q - W$
 $\Delta U = 400 \text{ J} - 150 \text{ J} = 250 \text{ J}$
The change in the internal energy of the gas is 250 J. This means the internal energy of the gas has increased.